Yan Wang

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EDUCATION

Wuhan University Wuhan, China
BS, Biological Science, GPA:3.91/4.0(2/33) *Sep.2023-Present*
Core course: Cell Biology(95) Molecular Biology(96) Genetics(96)
Biochemistry(95) Neurobiology(91) Gene Engineering(93)

Tsinghua University High School Beijing, China
National College Entrance Examination, Biology:100/100 *Sep.2020-Jun.2023*

SELF-INTRODUCTION

I have research experience in synthetic biology and biocomputation, biochemistry, and neurobiology, with the ability to independently carry out projects from conceptual design to experimental implementation and optimization.

I developed academic English proficiency through the Hongyi Honors Program at Wuhan University, where all major courses were delivered in English, as well as through training in an English-speaking laboratory environment (TOEFL: 106).

RESEARCH EXPERIENCE

Medical Research Institute, Wuhan University Wuhan, China
Joint lab of Prof.Yanxun Yu and Prof. Youngnam Jin *Jan.2025-Present*

Project title: Analysis of the functional sites of INS-1 neuropeptide in the alcohol perception neural circuit of C.elegans

- Identified that loss of INS-1 neuropeptide reverses ethanol avoidance behavior under high-salt conditions, indicating it's participation in integrating environmental information.
- Construct neuron-specific knockout/rescue INS-1 worm strains and conduct behavioral analysis.
- Use neuronal calcium imaging, optogenetic tools and confocal microscopy to explore the effect of ins-1 on neural circuit function.
- It has been identified that ASEL, AIY and AIZ serve as an important target for Ins-1 through my research.
- This project was carried out independently by me.

Project title: Performance Optimization of the Non-Diffusive Proximity Tagging System POST-IT Based on Directed Evolution

- The POST-IT system developed in our lab uses PafA ligase to label small drug target proteins.
- Use yeast display technology with random mutation to obtain a PafA mutant library.
- Obtain highly performing mutants through FACS.
- Initially, a mutant was obtained that significantly reduced the required labeling time and the substrate concentration.
- Detect the efficacy of the PafA mutant in the recognition of small molecule targets.

College of Life Science, Wuhan University Wuhan, China

Project for iGEM competition *Nov.2024-Oct.2025*

Project title: LOGIC -Lasting Optimized Genetic Input Calculator

- The core concepts, prototype design and technical map were proposed by me.
- Construct logic gates at the colony level using spatial diffusion and high-pass band-pass circuits.
- Build a series addition calculator using selected orthogonal quorum sensing systems.
- Separate the functional domain of the degradation enzyme and add a light-controlled degradation tag.

TECHNICAL SKILLS

- Microbial and model organism handling: *E. coli*, yeast, and *C. elegans*.
- Molecular cloning, protein expression and purification, Western blot, FACS, and routine biochemical assays, mammalian cell culture, etc. Microinjection, calcium imaging, optogenetic manipulation, and behavioral assays in *C. elegans*.
- C, HTML, CSS, etc. Familiar with basic single-cell RNA-seq data analysis workflows in R.
- Basic structural biology tools(AlphaFold and PyMOL), data analysis, data visualization.

ACADEMIC ACTIVITIES

Tsinghua University Beijing, China

National Forum on China's "Top-Tier Talent Program 2.0" in Biology *May.2025*

One of two delegates selected by Wuhan University.

Westlake University Hangzhou, China

"Frontiers of Life Science" International Undergraduate Summer School *Jul.2025-Aug.2025*

Beijing Bioaz Data Technology Co., Ltd. Beijing, China

Internship in AI-Driven Biological Research *Jan.2026*

HONORS & AWARDS

iGEM Foundation *Nov.2025*

Gold Medal, International Genetically Engineered Machine Competition (iGEM);

Core Team Member and Project Prototype Designer.

Chinese Society of Blotechnology *Aug.2025*

Gold Prize , White(Industry) Track, SynBio Challenges(an international contest on synthetic biology)

Silver Prize , Synthetic Cell Track, SynBio Challenges

Wuhan University

Liu Daoyu Creative Learning Scholarship(awarded to 36 students university-wide) *Jan.2026*

Hongyi Honors College's Annual Top Ten Scientific and Technological Advances Award *Oct.2025*

First-Class Outstanding Freshman Scholarship *Sep.2023*

Two years of Outstanding Student Scholarship

Two years of Outstanding Student

Two years of Outstanding Student Leader